Mini Project 2: Tight-binding model

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October 19, 2026

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1 Problem Statement

The electronic structures of two (imaginary) two-dimensional crystals, A and B, are described by a tight-binding model with a two-orbital basis. The dependence of the overlap and hopping parameters, η_{nm} and t_{nm} on the distance R between lattice sites is given by the following functions:

$$\eta_{11}(R) = \left(1 + \frac{R}{d_0} + \frac{R^2}{3d_0^2}\right) e^{-\frac{R}{d_0}}$$

$$\eta_{12}(R) = \frac{1}{2}\left(1 + \frac{R}{d_0} + \frac{R^2}{64d_0^2}\right) e^{-\frac{R}{2d_0}} - \frac{1}{2}e^{-\frac{R}{d_0}}$$

$$\eta_{22}(R) = \left(1 + \frac{R}{2d_0} + \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

$$t_{11}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}}$$

$$t_{12}(R) = -t_0\left(1 + \frac{2R}{d_0}\right) e^{-\frac{R}{d_0}} + t_0\left(1 + \frac{R}{16d_0}\right) e^{-\frac{R}{2d_0}}$$

$$t_{22}(R) = -t_0\left(1 + \frac{R}{2d_0} - \frac{R^2}{24d_0^2}\right) e^{-\frac{R}{2d_0}}$$

where d_0 and t_0 are positive constants that we use as length and energy units, respectively. The expressions for the hopping parameters are valid for $R > 3d_0$ and we have $t_{nm}(0) = 0$. The atomic energies associated with the two orbitals are $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$.

1. Calculate the electronic band structures of the two crystals assuming that the lattice constants is: (i) $a = 5d_0$; (ii) $a = 7d_0$; (iii) $a = 9d_0$. Include the nearest neighboor an the next nearest neighbor contributions.
2. Compare the results in (1) with the corresponding energy spectra calculated: (i) using only nearest neighbor contributions; (ii) neglecting the overlap integral, $\eta_{nm}(R) = \delta_{n,m}\delta_{R,0}$.

Note. The two crystals considered in this project have the following two-dimensional structures:

1. **Crystal A:** A square lattice with lattice constant a and primitive lattice vectors

$$\mathbf{a_1} = (a, 0), \qquad \mathbf{a_2} = (0, a).$$

   The primitive unit cell contains one lattice site at ($\boldsymbol{\tau}_1 = (0, 0)$), with two orbitals labeled 1 and 2.

2. **Crystal B:** A rectangular lattice with lattice constants a and $b = 1.5a$, and primitive lattice vectors

$$\mathbf{a}_1 = (a, 0), \qquad \mathbf{a}_2 = (0, b).$$

   The primitive unit cells contains a two-sites basis:

   - **Site 1:** $\boldsymbol{\tau}_1 = (0, 0)$, with orbitals 1 and 2.
   - **Site 2:** $\boldsymbol{\tau}_2 = (a/2, b/2)$, with orbital 1 only.

To skip the derivation and see the results directly, go to the last paragraph on page 6 for question 1, page 7 for question 2, and the second half of page 8 for question 3.

2 Electronic Band Structure

2.1 Crystal A

Using the prescribed hopping (t_{nm}) and overlap (η_{nm}) matrix elements, the electronic band structure is obtained by solving the generalized tight-binding eigenvalue problem

$$[\bar{\epsilon}\bar{\eta}(\mathbf{k}) + \bar{t}(\mathbf{k})]\, \mathbf{c}_\nu(\mathbf{k}) = E_\nu(\mathbf{k})\bar{\eta}(\mathbf{k})\mathbf{c}_\nu(\mathbf{k}), \tag{1}$$

where $\bar{\epsilon}$, $\bar{\eta}(\mathbf{k})$, and $\bar{t}(\mathbf{k})$ denote the on-site energy, overlap, and hopping matrices, respectively, while $E_\nu(\mathbf{k})$ represents the energy of band ν. Crystal A consists of a square lattice with two atomic orbitals per site, resulting in two electronic bands. The on-site energy matrix is

$$\bar{\epsilon} = \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix}, \tag{2}$$

with $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$. Including first- and second-nearest-neighbor interactions, the overlap and hopping matrices can be expressed as

$$\bar{\eta}(\mathbf{k}) = \begin{pmatrix} 1 + \eta_{11}(a) f_{\mathrm{NN}} + \eta_{11}(\sqrt{2}a) f_{\mathrm{NNN}} & \eta_{12}(a) f_{\mathrm{NN}} + \eta_{12}(\sqrt{2}a) f_{\mathrm{NNN}} \\ \eta_{21}(a) f_{\mathrm{NN}} + \eta_{21}(\sqrt{2}a) f_{\mathrm{NNN}} & 1 + \eta_{22}(a) f_{\mathrm{NN}} + \eta_{22}(\sqrt{2}a) f_{\mathrm{NNN}} \end{pmatrix}, \tag{3}$$

and

$$\bar{t}(\mathbf{k}) = \begin{pmatrix} t_{11}(a)f_{\mathrm{NN}} + t_{11}(\sqrt{2}a)f_{\mathrm{NNN}} & t_{12}(a)f_{\mathrm{NN}} + t_{12}(\sqrt{2}a)f_{\mathrm{NNN}} \\ t_{21}(a)f_{\mathrm{NN}} + t_{21}(\sqrt{2}a)f_{\mathrm{NNN}} & t_{22}(a)f_{\mathrm{NN}} + t_{22}(\sqrt{2}a)f_{\mathrm{NNN}} \end{pmatrix}, \tag{4}$$

where the nearest-neighbor (NN) and next-nearest-neighbor (NNN) structure factors are

$$f_{\mathrm{NN}}(\mathbf{k}) = 2\left[\cos(k_x a) + \cos(k_y a)\right], \tag{5}$$

$$f_{\mathrm{NNN}}(\mathbf{k}) = 4\cos(k_x a)\cos(k_y a). \tag{6}$$

The real-space square lattice and its corresponding reciprocal lattice are illustrated in Figs. 1(a) and (b),respectively. The electronic band structures were calculated along the high-symmetry path $\Gamma - X - M - \Gamma$ for three lattice constants: $a = 5d_0$, $7d_0$, and $9d_0$.

For $a = 5d_0$ [Fig. 1(c)], the calculated bands exhibit pronounced divergences. These features are associated with the singularity of the overlap matrix $\bar{\eta}(\mathbf{k})$, indicating a breakdown of the generalized eigenvalue description at this lattice spacing. Consequently, the divergent eigenvalues should not be interpreted as physically meaningful electronic energies within the present model.

Increasing the lattice constant to $a = 7d_0$ [Fig. 1(d)] produces a more regular band structure, with a clear separation between the two bands. At $a = 9d_0$ [Fig. 1(e)], the bands become progressively flatter, while their separation approaches the difference between the isolated-orbital energies.

This behavior follows from the exponential decay of the hopping and overlap matrix elements with interatomic distance. As the lattice expands, intersite coupling weakens, reducing the $\mathbf{k}$-dependence of the eigenvalues. In the limit of large interatomic separation, $\bar{t}(\mathbf{k}) \to 0$ and $\bar{\eta}(\mathbf{k}) \to I$, so the two bands approach the isolated-orbital energies $\epsilon_1 = 0$ and $\epsilon_2 = 0.8t_0$. The corresponding reduction in bandwidth reflects the increasing localization of the electronic states.

3 Crystal B: Centered Rectangular Lattice

Crystal B is modeled on a centered rectangular lattice with a two-site basis:

- Site 1 at $\vec{\tau}_1 = (0, 0)$, which has two orbitals (orbital 1 and orbital 2)
- Site 2 at $\vec{\tau}_2 = \left(\frac{a}{2}, \frac{b}{2}\right)$, which has one orbital (orbital 1)

This gives a 3-orbital basis:

1. Orbital 1 on Site 1 ($\epsilon_1 = 0$)
2. Orbital 2 on Site 1 ($\epsilon_2 = 0.8t_0$)
3. Orbital 1 on Site 2 ($\epsilon_3 = \epsilon_1 = 0$)

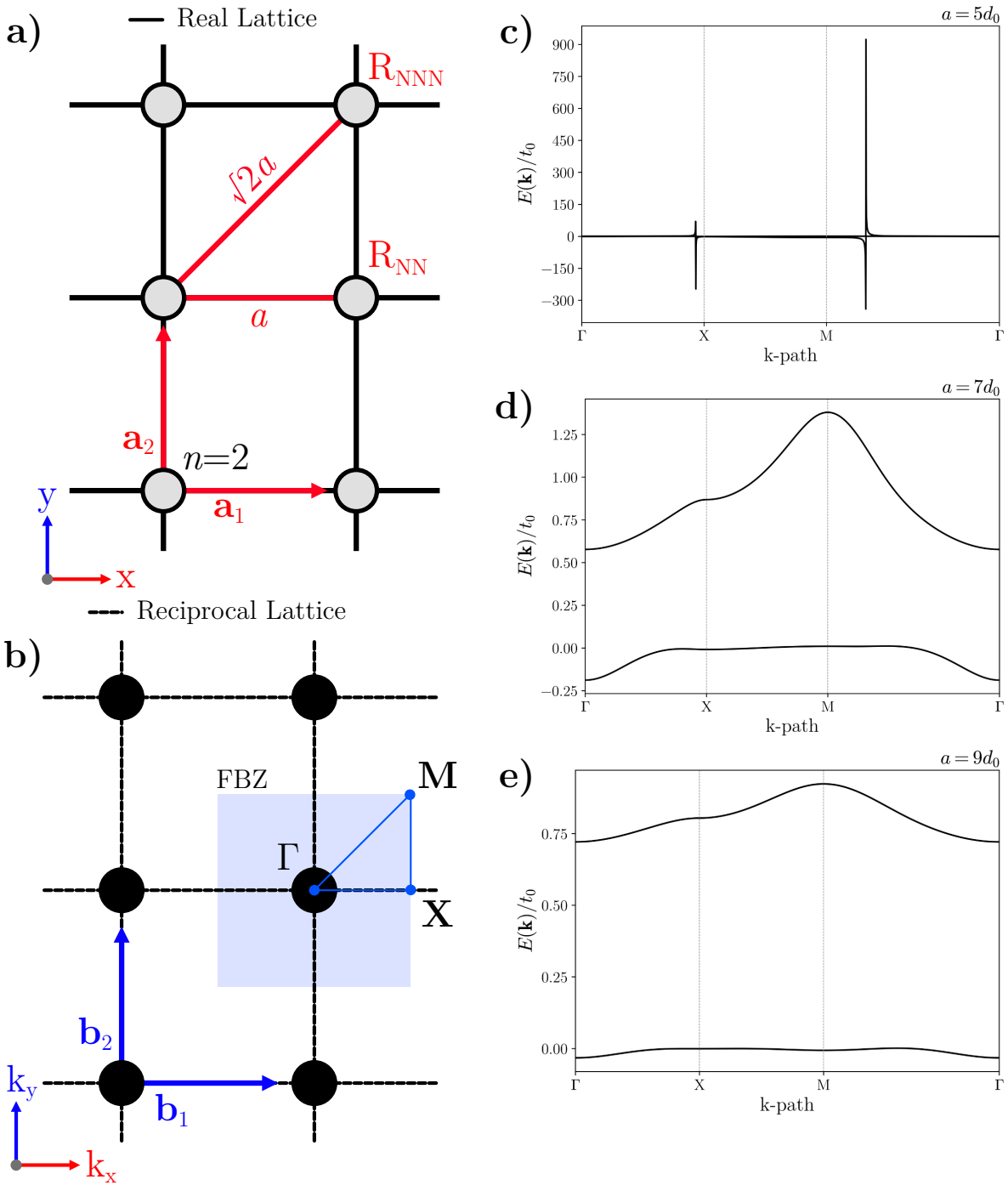


Figure 1: (a) Real-space square lattice of crystal A, indicating the lattice vectors and nearest-neighbor distances. (b) Corresponding reciprocal lattice, first Brillouin zone, and high-symmetry points. Electronic band structures along $\Gamma - X - M - \Gamma$ for (c) $a = 5d_0$, (d) $a = 7d_0$, and (e) $a = 9d_0$.

Since State 1 is orbital 1 (Site 1), State 2 is orbital 2 (Site 1), and State 3 is orbital 1 (Site 2), the off-diagonal inter-sublattice couplings map directly to the radial functions:

$$\eta_{13}(R) \equiv \eta_{11}(R), \qquad t_{13}(R) \equiv t_{11}(R)$$
$$\eta_{23}(R) \equiv \eta_{12}(R), \qquad t_{23}(R) \equiv t_{12}(R)$$

Therefore, the eigenvalue problem as a function of $\vec{k}$ is 3×3, and $\bar{\epsilon}$ is given by:

$$\bar{\epsilon} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0.8t_0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with $\bar{t}$ and $\bar{\eta}$ defined as 3×3 Hermitian matrices:

$$\bar{t}(\vec{k}) = \begin{pmatrix} t_{11}(\vec{k}) & t_{12}(\vec{k}) & t_{13}(\vec{k}) \\ t_{12}^*(\vec{k}) & t_{22}(\vec{k}) & t_{23}(\vec{k}) \\ t_{13}^*(\vec{k}) & t_{23}^*(\vec{k}) & t_{33}(\vec{k}) \end{pmatrix}, \qquad \bar{\eta}(\vec{k}) = \begin{pmatrix} \eta_{11}(\vec{k}) & \eta_{12}(\vec{k}) & \eta_{13}(\vec{k}) \\ \eta_{12}^*(\vec{k}) & \eta_{22}(\vec{k}) & \eta_{23}(\vec{k}) \\ \eta_{13}^*(\vec{k}) & \eta_{23}^*(\vec{k}) & \eta_{33}(\vec{k}) \end{pmatrix}$$

For this lattice, contributions are classified by the actual physical interatomic distance $R = |\vec{R}_j + (\vec{\tau}_m - \vec{\tau}_n)|$ into on-site, nearest-neighbor, and next-nearest-neighbor shells.

1. On-Site ($R = 0$) Contribution

Because Site 1 at $\vec{\tau}_1 = (0, 0)$ and Site 2 at $\vec{\tau}_2 = \left(\frac{a}{2}, \frac{b}{2}\right)$ are physically distinct atoms, there is no $R = 0$ inter-sublattice term ($\eta_{13}^{(0)} = \eta_{23}^{(0)} = 0$ and $t_{nm}^{(0)} = 0$). The on-site overlap matrix is purely diagonal:

$$\eta_{nm}^{(0)}(\vec{k}) = \eta_{nm}(0) = \delta_{nm}, \qquad t_{nm}^{(0)}(\vec{k}) = 0$$

Specifically:

$$\eta_{11}^{(0)} = \eta_{22}^{(0)} = \eta_{33}^{(0)} = 1, \qquad \eta_{12}^{(0)} = \eta_{13}^{(0)} = \eta_{23}^{(0)} = 0$$

2. Nearest Neighbor Contributions '(1)'

Intra-Sublattice Elements ($11, 22, 33$ and 12)

For the intra-sublattice terms ($\vec{\tau}_m - \vec{\tau}_n = \vec{0}$), the nearest neighbors on the Bravais lattice lie along the axes at $\vec{R}_j = (\pm a, 0)$ and $(0, \pm b)$ with distances $R = a$ and $R = b$, respectively.

For the diagonal elements:

$$\eta_{nn}^{(1)}(\vec{k}) = 2\eta_{nn}(a)\cos(k_x a) + 2\eta_{nn}(b)\cos(k_y b)$$
$$t_{nn}^{(1)}(\vec{k}) = 2t_{nn}(a)\cos(k_x a) + 2t_{nn}(b)\cos(k_y b)$$

where for State 3 (orbital 1 on Site 2), $\eta_{33}^{(1)} = \eta_{11}^{(1)}$ and $t_{33}^{(1)} = t_{11}^{(1)}$.

For the off-diagonal intra-sublattice elements (η_{12}, t_{12}), since orbitals 1 and 2 reside on the same site, the on-site overlap is zero by orthogonality. The nearest-neighbor interaction yields:

$$\eta_{12}^{(1)}(\vec{k}) = 2\eta_{12}(a)\cos(k_x a) + 2\eta_{12}(b)\cos(k_y b)$$
$$t_{12}^{(1)}(\vec{k}) = 2t_{12}(a)\cos(k_x a) + 2t_{12}(b)\cos(k_y b)$$

Inter-Sublattice Elements (η_{13}, t_{13} and η_{23}, t_{23})

For the inter-sublattice couplings, $\vec{\tau}_2 - \vec{\tau}_1 = \left(\frac{a}{2}, \frac{b}{2}\right)$. The four nearest-neighbor Site 2 atoms surrounding Site 1 lie symmetrically at displacements $\vec{d} = \left(\pm\frac{a}{2}, \pm\frac{b}{2}\right)$, all at the exact same physical distance $R = \frac{1}{2}\sqrt{a^2+b^2}$. These correspond to the four Bravais lattice vectors:

$$\vec{R}_j \in \{(0,0), (-a,0), (0,-b), (-a,-b)\}$$

$$\begin{aligned}\eta_{13}^{(1)}(\vec{k}) &= \eta_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left[1 + e^{-iak_x} + e^{-ibk_y} + e^{-i(ak_x+bk_y)}\right] \\ &= \eta_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)\end{aligned}$$

$$t_{13}^{(1)}(\vec{k}) = t_{13}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

$$\eta_{23}^{(1)}(\vec{k}) = \eta_{23}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

$$t_{23}^{(1)}(\vec{k}) = t_{23}\left(\frac{\sqrt{a^2+b^2}}{2}\right)\left(1 + e^{-iak_x}\right)\left(1 + e^{-ibk_y}\right)$$

3. Next-Nearest Neighbor Contributions '(2)'

Intra-Sublattice Elements ($11, 22, 33$ and 12)

Displacements to the four corner unit cells lie at $\vec{R}_j = (\pm a, \pm b)$ with distance $R = \sqrt{a^2+b^2}$:

$$\eta_{nn}^{(2)}(\vec{k}) = 4\eta_{nn}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$
$$t_{nn}^{(2)}(\vec{k}) = 4t_{nn}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$

$$\eta_{12}^{(2)}(\vec{k}) = 4\eta_{12}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$
$$t_{12}^{(2)}(\vec{k}) = 4t_{12}\left(\sqrt{a^2+b^2}\right)\cos(ak_x)\cos(bk_y)$$

Inter-Sublattice Elements (η_{13}, t_{13} and η_{23}, t_{23})

The second inter-sublattice shell consists of 8 neighbors formed by:

- 4 neighbors extended along x at displacements $\vec{d} = \left(\pm\frac{3a}{2}, \pm\frac{b}{2}\right)$ with distance $R = \frac{1}{2}\sqrt{9a^2 + b^2}$, generated by $\vec{R}_j \in \{(a, 0), (a, -b), (-2a, 0), (-2a, -b)\}$.
- 4 neighbors extended along y at displacements $\vec{d} = \left(\pm\frac{a}{2}, \pm\frac{3b}{2}\right)$ with distance $R = \frac{1}{2}\sqrt{a^2 + 9b^2}$, generated by $\vec{R}_j \in \{(0, b), (-a, b), (0, -2b), (-a, -2b)\}$.

$$\eta_{13}^{(2)}(\vec{k}) = \eta_{13}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x - bk_y)} + e^{-2iak_x} + e^{-i(2ak_x + bk_y)}\right] + \eta_{13}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x + bk_y)} + e^{-2ibk_y} + e^{-i(ak_x + 2bk_y)}\right]$$

$$t_{13}^{(2)}(\vec{k}) = t_{13}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x - bk_y)} + e^{-2iak_x} + e^{-i(2ak_x + bk_y)}\right] + t_{13}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x + bk_y)} + e^{-2ibk_y} + e^{-i(ak_x + 2bk_y)}\right]$$

$$\eta_{23}^{(2)}(\vec{k}) = \eta_{23}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x - bk_y)} + e^{-2iak_x} + e^{-i(2ak_x + bk_y)}\right] + \eta_{23}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x + bk_y)} + e^{-2ibk_y} + e^{-i(ak_x + 2bk_y)}\right]$$

$$t_{23}^{(2)}(\vec{k}) = t_{23}\left(\frac{\sqrt{9a^2+b^2}}{2}\right)\left[e^{iak_x} + e^{i(ak_x - bk_y)} + e^{-2iak_x} + e^{-i(2ak_x + bk_y)}\right] + t_{23}\left(\frac{\sqrt{a^2+9b^2}}{2}\right)\left[e^{ibk_y} + e^{i(-ak_x + bk_y)} + e^{-2ibk_y} + e^{-i(ak_x + 2bk_y)}\right]$$

The lower-triangle elements are given by the Hermitian conjugate relations:

$$t_{ji}(\vec{k}) = t_{ij}^*(\vec{k}), \qquad \eta_{ji}(\vec{k}) = \eta_{ij}^*(\vec{k})$$

4. Total Matrix Assembly & Approximations

The total matrix elements are combined as:

$$\eta_{nm}(\vec{k}) = \eta_{nm}^{(0)}(\vec{k}) + \eta_{nm}^{(1)}(\vec{k}) + \eta_{nm}^{(2)}(\vec{k})$$
$$t_{nm}(\vec{k}) = t_{nm}^{(0)}(\vec{k}) + t_{nm}^{(1)}(\vec{k}) + t_{nm}^{(2)}(\vec{k})$$

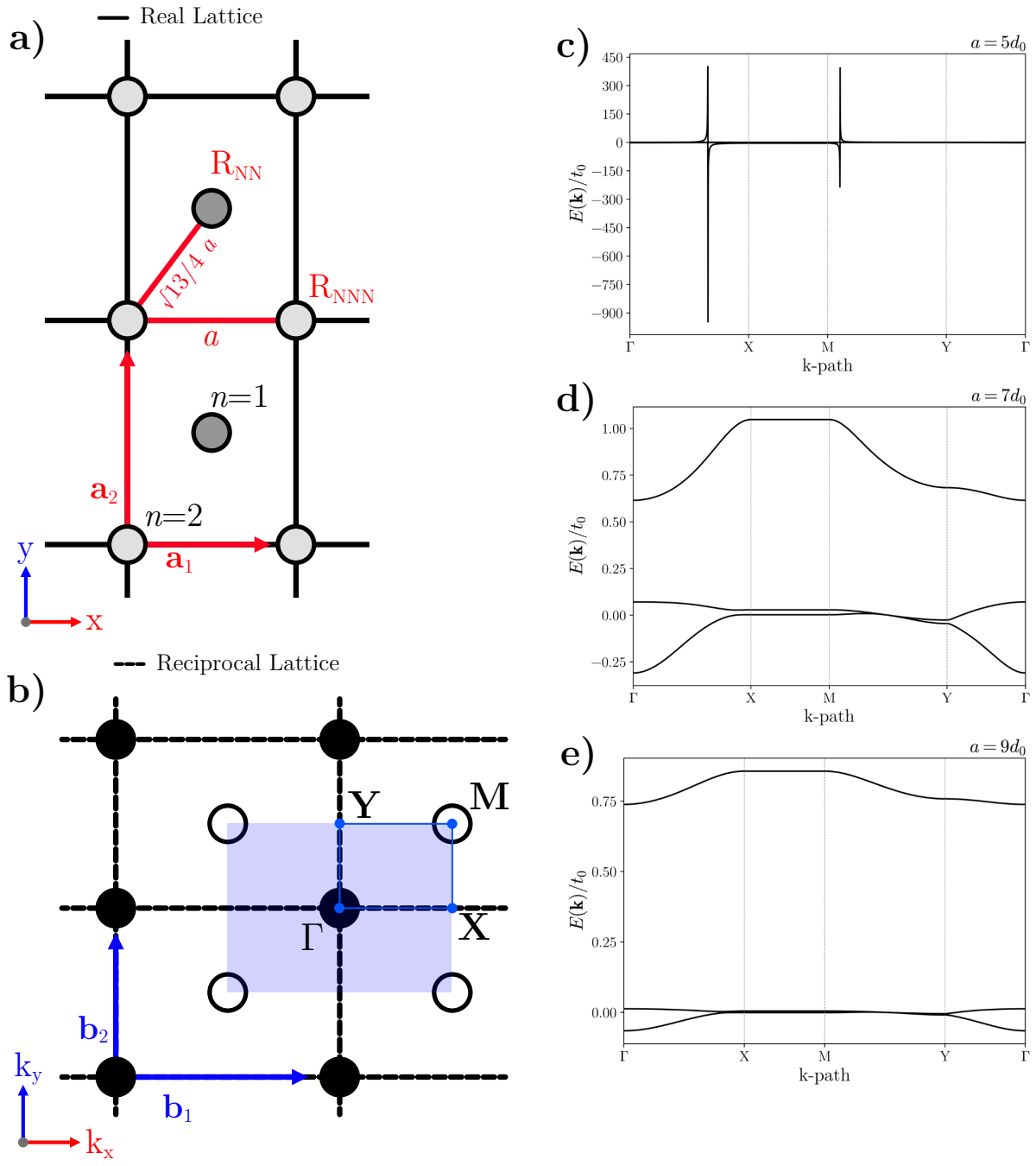


Figure 2: Caption

4

5 References

Girvin, S. M., and Yang, K., Modern Condensed Matter Physics (Cambridge University Press, Cambridge, 2019).